Tutorial – Week 1: Essence and Accidents of Software Engineering

1. Categorize each as either an Essential Difficulty or an Accidental Difficulty, and provide a brief justification
    - Determining the interlocking business rules for a cross-border tax calculation engine.
    - Configuring the YAML files for a Kubernetes cluster to ensure high availability.
    - Resolving a "Segmentation Fault" caused by manual memory management in C++.
    - Mapping out the conflicting requirements between three different stakeholders for a new hospital management system.
    - Optimizing the syntax of a Python script to run faster on a specific legacy processor.

2. What is software crisis?

3. Define essential difficulties and accidental difficulties in the context of software engineering. Provide two examples of each type of difficulty.

4. Software complexity leads to many difficulties, what are they?

5. In your opinion, why is software perceived as the most conformable element?

6. Brooks suggests that software development is inherently hard, and advances in programming languages, tools, or development environments can only offer limited improvements in productivity. Evaluate this claim in the context of modern software engineering practices and technologies.

7. Is there any possible "silver bullet"? What are they?

8. How do you summarize the essay?

9. Discuss how contemporary practices like Agile and AI-enhanced tools are used to tackle essential difficulties identified by Fredrick Brooks.